A SMART AI FRAMEWORK FOR AUTOMATED CIVIL COMPLAINT MANAGEMENT AND CITIZEN ENGAGEMENT IN SRI LANKA

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
D.M.C. Shehara	IT22593900	

The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor: Date

……..
……
Signature of the co-supervisor: Date

ABSTRACT

E-governance systems play a vital role in improving communication between citizens and government institutions by enabling efficient handling of public complaints. However, in many developing countries, including Sri Lanka, complaint management processes are still largely manual, leading to delays, misclassification, and reduced service efficiency. Therefore, there is a need for an intelligent system that can automate complaint classification and routing while improving overall service delivery.

This research proposes GovConnect, a web-based government–citizen communication platform that integrates Artificial Intelligence techniques to enhance complaint management. The system utilizes Retrieval-Augmented Generation (RAG) combined with the LLaMA 3 Large Language Model (LLM) to classify citizen complaints into relevant government departments based on contextual understanding. A structured knowledge base was developed using data collected from government offices, including department descriptions, keywords, synonyms, and complaint examples.

The methodology includes data collection, preprocessing, embedding generation using nomic-embed-text, and classification through a hybrid approach combining semantic similarity, keyword matching, and a confidence threshold mechanism. The system also supports a complete workflow where complaints are routed to ministers and assigned to agents based on department and geographical location. The implementation was carried out using the MERN stack with JWT-based authentication and role-based access control.

Experimental testing was conducted using real-world complaint data, where the system achieved a classification accuracy of 87.18%, correctly classifying 34 out of 39 test cases. The results demonstrate that the proposed approach effectively handles unstructured complaint data and improves classification performance compared to traditional methods.

In conclusion, GovConnect provides a scalable and intelligent solution for automating complaint management, improving efficiency, accuracy, and transparency in government–citizen interactions.

Keywords: E-governance, Complaint Classification, RAG, LLaMA, Natural Language Processing

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LIST OF ABBSERVATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DB	Database
G2C	Government-to-Citizen
GovConnect	Government Connect Platform
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
LLaMA	Large Language Model Meta AI
MERN	MongoDB, Express.js, React.js, Node.js
ML	Machine Learning
NLP	Natural Language Processing
NFR	Non-Functional Requirement
ODM	Object Data Modeling
OOP	Object-Oriented Programming
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
REST	Representational State Transfer
TF-IDF	Term Frequency–Inverse Document Frequency
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator

1. INTRODUCTION

In developing countries, such as Sri Lanka, the interaction between public authorities and citizens is a crucial element in the delivery of public services. Yet, conventional approaches to citizen complaint management are often archaic, manual and inefficient. This can result in time delays, incorrect classification, and failure to prioritise urgent issues. The manual handling of complaints becomes a major issue with increasing numbers of complaints, and the unstructured manner in which complaints are entered, often in Sinhalese and English, poses a challenge, particularly in rural regions with limited digital literacy. The current complaint management systems do not offer timely responses and solutions to pressing issues. Rather, they are dealt with on a first come, first serve basis. This failure to adopt an automated structured approach not only hampers the efficiency of the government but also reduces citizens' confidence in the government's responsiveness. This study introduces GovConnect, a cutting-edge AI-based platform aimed at transforming the complaint management system in Sri Lanka. The system leverages new technologies such as Natural Language Processing (NLP) and Machine Learning (ML) to automate complaint categorization, prioritisation, and allocation. The smart system will automatically classify complaints into the appropriate department based on the content of the complaint, prioritize the complaint based on urgency and route them for immediate resolution. GovConnect has a number of innovations. First, AI-Driven Classification classifies complaints made by citizens to the correct government department based on NLP methods, even if they are in informal or multilingual text. Second, Dynamic Prioritization takes into account the feedback from our community through a voting system, enabling the community to put priority levels on issues. Issues are then dynamically prioritised based on this feedback and urgency models. Third, Real-Time Tracking provides real-time monitoring for government officers (ministers and agents). Ministers have the ability to view complaints, allocate them to agents based on the department and district of the citizen, and monitor the progress of complaints. And finally,

Transparency and Accountability are enhanced by giving citizens real-time information on the status of their complaints, which promotes transparency and accountability in government. The main aim of GovConnect is to build a more efficient, transparent, and citizen-friendly complaint resolution system. GovConnect uses AI and automation to improve the efficiency, effectiveness and equity of the government's service delivery model, making the complaint resolution faster, more accurate and fair. Not only are citizens not just service consumers but involved in the decision-making process, enhancing government responsiveness. This work shows that an AI-based solution can overcome the communication barrier between citizens and government, enabling a more efficient and participatory government. The integration of AI-based automation, real-time data access, and citizen participation holds the potential to revolutionise Sri Lanka's complaints system and thereby provide more responsive, transparent, and accountable services to citizens.

1.1.Background & Literature Review

Artificial Intelligence (AI) is becoming a key component of e-governance systems, particularly in enhancing government-to-citizen (G2C) communication and complaint management. In many countries, citizen complaints are still processed manually or semi-electronically through reading a complaint description, determining the responsible department, routing the complaint, and then tracking its status. This can be a slow, tedious, and inefficient process for handling large numbers of complaints. This can impact service quality and citizen trust. Thus, smart complaint management systems are required to automate the process of understanding, classifying, prioritising and allocating complaints. In this study, we build a web-based government-to-citizen communication system. The system enables citizens to lodge complaints online, automatically identifies the responsible government department using a Retrieval-Augmented Generation (RAG) system with an updated Ollama LLaMA 3 model, prioritises complaints using a machine learning model, and facilitates monitoring by ministers and agent assignment,

based on district and department relevance. Previous research indicates that various researchers have attempted to classify complaints using text mining, natural language processing (NLP) and machine learning (ML) methods. The initial research showed complaint reports can be automatically classified by text analysis. Sano, Yamaguchi, and Mine (2015) proposed an automatic classification system for city park complaints, and demonstrated that text classification can help local governments through administrative automation [1]. Fauzi (2018) developed an automatic complaint classification system based on classifier ensembles and confirmed that an ensemble of classifiers can outperform individual classifiers [2]. Qurat-ul-ain, Shaukat and Saif (2022) also proposed an NLP-based classification system and found that complaint texts can be used to automatically classify complaints [3]. These studies demonstrate that there are useful information in complaint text that can be used to aid decision-making in service systems. Subsequently, other works proposed approaches using machine learning and deep learning. Naik et al. (2023) compared consumer complaint classification using machine learning and deep learning and highlighted that an automated system helps increase consistency and ease of work [4]. Saha et al. (2024) developed an automated complaint classification and routing framework based on NLP and machine learning, noting that classification and routing are equally important as complaints need to reach the right departments for effective action [5]. But many of these previous approaches rely on classical text feature engineering techniques such as Bag-of-Words and TF-IDF, which do not capture semantic meaning and relationships in complaint text. These methods might not work well on short, ambiguous, and stylistically diverse complaints. Developments in deep learning have led to more advanced models for this task. Recurrent neural networks (RNN) like Long Short-Term Memory (LSTM) were designed to learn long-term dependencies and resolve issues with traditional RNNs [6]. Subsequently, Vaswani et al.'s Transformer architecture (2017) enhanced text understanding using self-attention, which allows for better capturing of text semantics [7]. The fine-tuning of large pretrained language models for text classification was also a substantial advance. Howard and

Ruder (2018) demonstrated the effectiveness of fine-tuning language models for improving text classification performance even with a small number of training examples [8]. Moreover, survey papers also show that deep learning approaches are superior to traditional machine learning algorithms in many text classification tasks because they can learn complex and informative features from text automatically [9]. The use of embeddings and Large Language Models (LLMs) has also been shown to improve complaint classification in recent studies. Rakhimzhanov, Belginova, and Yedilkhan (2025) suggested classifying public transport complaints using text mining and LLMs with embeddings, demonstrating that current semantic representations can help to better understand complaints in public services [10]. This is particularly important for e-governance platforms because citizen complaints might involve multiple problems, implicit remarks and technical terms. LLMs are more likely to understand the meaning of complaints, beyond basic keyword matching, than previous methods. Retrieval-Augmented Generation (RAG) is another key development. RAG was developed by Lewis et al. (2020) and combines the use of external knowledge retrieval and language model generation to enhance performance on knowledge-rich NLP tasks [11]. Recently, Adaptive-RAG showed how retrieval can be used adaptively based on the difficulty of the input, improving efficiency and accuracy This is especially useful for complaint classification as words can represent different departments in different contexts. For instance, the words road, accident, hospital, garbage, or power cut may require context to be understood so that the complaint can be classified to the appropriate department. A RAG-based model can provide more accurate department classification than a pure text-based classification model by retrieving department descriptions, keywords, synonyms, and complaint examples from a knowledge base. While previous research has achieved promising results for complaint classification, there are a few drawbacks. Current studies primarily focus on classification performance and lack a full process for handling complaints for government. In practice, government complaint handling not only focuses on matching departmental

responsibilities, but also sorting complaints by urgency, routing them to the right office and personnel, allocating complaints to service agents, tracking them until the complaints are resolved. Current systems may not adequately address these considerations. Furthermore, many existing studies are not tailored to government complaint systems where complaints need to be associated with departments, ministers, districts and agents in a certain administrative workflow.

Thus, this work shows a potential solution by providing an integrated web-based complaint management system for citizen communication. In our system, citizens can lodge a complaint, the complaint is automatically classified into the right department with a RAG-based LLaMA 3 model, then prioritized with a RandomForestRegressor-based urgency model that takes into account both the priority of the complaint and voting by the public, and finally, the complaint is reviewed by the relevant minister, who assigns the complaint to the most appropriate agent in the district and department. Agents also update the statuses of the complaints (pending, in progress, done or rejected) and citizens and ministers can check the statuses through the system. This research differs from many previous studies which only consider classification based AI, but instead looks at a combination of AI-based complaint classification, urgency prediction, and real-world complaint assignment methods in an e-governance system. Thus, the proposed system stands out academically and practically through its advancement from isolated text classification towards an integrated intelligent complaint management system in public administration.

1.2Research gap

Artificial Intelligence (AI) is increasingly being used in e-governance systems and in particular, to improve government-to-citizen communication and

complaint management. In many developing countries, complaints made by citizens are still processed using manual or semi-digital processes, where officers read the complaint descriptions, determine the department responsible, and route the complaint; and later track its status. This can be time-consuming, uneven and inefficient, especially when dealing with a large number of complaints. Slow complaint classification and assignment can lead to lower service quality and customer satisfaction. Hence, complaint management systems with intelligence are required to automatically understand, classify, prioritise and assign complaints. In this study, we develop an online government-to-citizen communication platform. The system enables citizens to lodge complaints via the web, automatically determines the concerned government department through a Retrieval-Augmented Generation (RAG) system with a fine-tuned Ollama LLaMA 3 model, prioritises complaints through a machine learning model, and facilitates ministerial monitoring and an agent assignment system based on district and department relevance.

Previous research shows that several studies have investigated complaint classification using text mining, natural language processing and machine learning. Previous studies demonstrated the ability to automatically classify complaint reports using text analysis. Sano, Yamaguchi, and Mine (2015) proposed an automatic classification system for city park complaints and confirmed that text categorization can help local governments by automating processes [1]. Fauzi (2018) introduced an automatic complaint classification system based on classifier ensembles and showed that the ensemble of multiple classifiers can increase accuracy and robustness [2]. Qurat-ul-ain, Shaukat, and Saif (2022) also proposed a natural language processing (NLP) approach to complaint classification, and confirmed that complaint descriptions can be used to classify complaints automatically [3]. These studies demonstrate that complaint descriptions can be leveraged to inform service system decision-making.

Subsequently, studies applied machine learning and deep learning methods. Naik et al. (2023) built on these works by examining consumer complaint

classification using machine learning and deep learning and found that the automation systems can be consistent and reduce human effort [4]. Saha et al. (2024) developed an automated complaint classification and routing system using NLP and machine learning to identify the importance of correct classification and correct routing since complaints need to be viewed by the right authority to be effective [5]. But many of these previous approaches use conventional feature extraction techniques such as Bag-of-Words and TF-IDF, which are limited in capturing the semantic meaning and context of complains. These methods may not work well for short, ambiguous or stylistically diverse complaints.

As deep learning technology progresses, new models have emerged for text classification. Long Short-Term Memory (LSTM) networks and other recurrent neural networks were developed to model long-term dependencies, and address issues in conventional recurrent networks [6]. Subsequently, the Transformer introduced by Vaswani et al. (2017) greatly enhanced contextual representations of text with self-attention, allowing it to better understand semantic relations across words [7]. The use of fine-tuning large language models for text classification also emerged. Howard and Ruder (2018) demonstrated that fine-tuning language models can lead to improved classification accuracy even with small amounts of labeled data [8]. Furthermore, survey papers also show that deep learning approaches outperform traditional machine learning approaches in many text classification tasks, as they can automatically learn rich and useful representations of text [9]

Recent studies have also improved complaint classification with embeddings and Large Language Models (LLMs). Rakhimzhanov, Belginova, and Yedilkhan (2025) developed an automated classification system for public transport complaints using text mining and LLMs with embeddings, demonstrating that current semantic representations can enhance the understanding of complaints for public services [10]. This is crucial for e-governance systems, as citizen complaints may encompass multiple issues, indirect statements and technical terms. LLM-based methods are more adept

at capturing meaning than previous methods.

An additional key development is Retrieval-Augmented Generation (RAG). The RAG approach, proposed by Lewis et al. (2020) retrieves relevant information from an external source and uses a language model to generate text for knowledge-intensive NLP tasks [11]. Recently, Adaptive-RAG showed that the search strategy can be dynamically changed based on the difficulty of the input prompt to achieve greater efficiency and accuracy [12]. This is important for complaint classification as the same term may suggest different departments. For instance, the words road, accident, hospital, garbage, or power cut may require context for the complaint to be classified into the proper department. Through a structured knowledge base, the RAG-based model can retrieve department descriptions, keywords, synonyms, and examples of complaints, and generate more accurate and reliable classifications than a text classifier alone.

While prior research has achieved good results in complaint classification, there are some limitations. Most research focuses solely on accuracy, and does not offer a comprehensive solution for government complaint management. In practice, complaint handling in public service systems requires not only determining the correct department, but also sorting and prioritizing complaints, sending them to the appropriate authorities, allocating them to field agents, and tracking them until they are resolved. Current systems may not integrate these processes. Moreover, previous research is not tailored for government complaint systems that require that complaints be linked to departments, ministers, districts and agents in an established administrativeworkflow.

As a result, this paper fills this gap by offering an integrated complaint management solution in the form of a web application for government citizen communication. In our system, citizens can lodge complaints via the system, the complaint is automatically classified into the appropriate department using a RAG-based LLaMA 3 model, then prioritised using an urgency model based on RandomForestRegressor which takes into account the priority of the complaint and public voting, and then reviewed by the appropriate minister

who assigns the complaint to the most suitable agent in the relevant district and department. Agents can update the status of complaints (pending, in progress, done, and rejected) and citizens and ministers can track the complaint status. This study differs from many existing studies that only deal with classification, by integrating the use of AI-based complaint classification, urgency prediction, and practical assignment in an e-governance system. Thus, the proposed system is academically and practically significant as it is moving towards the development of a comprehensive intelligent complaint management system for public administration rather than just a simple text classification.

Reference	Functionalities of the Application	Technology and techniques Used for implementation	Platform/Usage	Accuracy
Traditional Government Complaint Systems	Manual complaint submission, manual classification, manual assignment	Human-based decision making	Web / Office-based	Low (time-consuming, error-prone)
Basic E-Government Portals	Online complaint submission, basic tracking	Rule-based systems, keyword matching	Web platforms	Low to Medium
Sano et al. [1]	Automatic classification of complaint reports	Text mining, basic NLP	Research prototype	Medium
Fauzi [2]	Complaint classification using ensemble classifiers	Machine Learning	Experimental system	Medium
Qurat-ul-ain et al. [3]	NLP-based complaint classification	NLP techniques	Research system	Medium
Naik et al. [4]	Consumer complaint classification	ML + Deep Learning models	Experimental datasets	Medium to High
Saha et al. [5]	Automated complaint classification and routing	NLP + Machine Learning	Prototype system	
Rakhimzhanov et al. [6]	Transport complaint classification using	LLM + Embeddings	Research system	High

	embeddings			
Existing Smart Complaint Apps	Complaint submission and tracking only	Basic database + UI	Mobile / Web apps	Medium
LLM-based Systems (General)	Text understanding and classification	Transformers, LLMs	Research / APIs	High but lacks domain knowledge
RAG-based Systems [7][8]	Knowledge-enhanced text processing	LLM + Retrieval-Augmented Generation	Advanced research	High
Chen et al. [16]	Predicts and analyzes government complaints using data-driven approaches	Machine Learning, Predictive Analytics	Web-based systems	High prediction accuracy
Gupta et al. [17]	Complaint lodging and tracking across multiple departments	AI-based grievance system, database management	Web & Mobile	Improved efficiency
Kusumawardani et al. [5]	Classification of public complaints into categories	Text classification, NLP techniques	Web-based	Medium
Zaidi et al. [22]	Automated support ticket routing and solution suggestion	Machine Learning, Intelligent systems	Web systems	High
Esperança et al. [13]	Proactive complaint management using AI	Semantic analysis, pattern recognition	Web systems	High
GovConnect – Proposed System	Complaint submission, auto classification, department suggestion, minister review, agent assignment, real-time tracking	RAG + LLaMA 3 (LLM),	Web-based real-world system	Higher accuracy + full workflow integration

1.3. Research Problem

The primary research question in this research is related to the inefficient, inconsistent and non-transparent nature of existing government-citizen complaint

management systems. Current systems in Sri Lanka and many developed countries have a high degree of manual processing based on human judgment, for the submission, classification, assignment and tracking of complaints. Such systems face a number of problems, most notably the unstructured data. Complaints are usually entered as unstructured text, which can be vague, informal, and/or incomplete. This poses a challenge for current systems to classify and route these complaints to the right department or agent, resulting in delays and mis-handling.

The second problem is the ineffectiveness of existing classification and prioritization processes. Existing systems may use basic rule-based classification or even manual interaction, which can be inaccurate and inconsistent. Such approaches are not integrated with prioritization, an essential element that guarantees fairness and timeliness of complaint handling. In reality, complaints need to be both classified and prioritised based on urgency. The absence of integration between classification and prioritization is a critical issue.

Moreover, the existing systems do not include citizen involvement in decision-making. Recent research shows the role of artificial intelligence in government administration and governance, but the current systems do not consider citizen feedback such as voting for prioritization. This has a negative impact on the responsiveness of government services and does not reflect the societal relevance of the complaint. By including citizen voting, we can adopt a more democratic and citizen-centred approach to complaint resolution, prioritising complaints with greater societal importance.

Manual allocation of complaints to departments and agents is another area of concern. Complaints are often manually assigned, without smart support, resulting in inefficiencies. Also, the absence of real-time monitoring and workflows makes it challenging for government departments to manage complaints. The lack of an integrated workflow that can manage classification, prioritisation and assignment in a holistic way results in disjointed processes and inefficient resource allocation.

Moreover, current systems lack scalability, particularly when dealing with high volumes of complaints. Manual workflows make it more challenging for government agencies to process complaints effectively. This could lead to a delayed complaint resolution process and, as a consequence, lower citizen satisfaction and trust. The need for a fast, efficient and accurate system to handle complaints which can be monitored in real-time and used by decision-makers is apparent.

To address these issues, this research recommends the design of a smart web-based

government-to-citizen platform, GovConnect. The system will use modern technologies such as Retrieval-Augmented Generation (RAG) and fine-tuned Large Language Models (LLaMA 3) to classify complaints into the appropriate department. It will also prioritise complaints based on citizen priority and public participation using machine learning models. The system will also enable monitoring at the ministerial level, as well as the assignment of intelligent agents based on department and district, enabling timely complaint resolution

Thus, the research problem can be defined as the absence of an automated, smart and integrated system to automatically and accurately classify, prioritise and allocate citizen complaints to the appropriate agents in government-citizen communication systems. The existing system in Sri Lanka and other parts of the world is inefficient, inaccurate and non-transparent, and it is heavily dependent on human interventions, which can result in errors, inconsistencies and opacity. Complaints are usually submitted through traditional or informal channels and are manually screened, classified and routed to departments. This is inefficient and error-prone and leads to routing problems. Urgent complaints that are important for society may not be processed in a timely manner and vice versa.

The first research question is how to correctly classify citizens' complaints to the right government department, as citizens may use different terms and descriptions. For instance, financial fraud may be referred to as 'theft' by some citizens, and as 'misappropriation' by others. This may result in manual classification errors. The challenge is how to use smart algorithms to decide which department is best to handle a particular complaint

The second problem relates to the creation of a prioritisation approach for complaints in an analytical and equitable way. Existing approaches may use first-in-best-served or a police officer's subjective assessment of urgency. The goal is to design a machine learning algorithm to produce urgency scores and prioritise from highest to lowest (priority), based on factors such as citizen feedback and voting.

The third research issue is to design a workflow that encompasses classification and prioritisation stages. The algorithm should not only classify but also prioritise the complaints. Furthermore, the prioritized and classified complaints need to be made available to officials. This involves making sure complaints are directed to the right departments, viewed by ministers and assigned to the right agents based on department and district

The problem is the absence of instant monitoring and tracking systems. Once the

complaints are allocated to agents, it is essential to have a system that supports agents to update complaint status (e.g. pending, in progress, completed, rejected) and ministers and citizens to monitor the status of complaints. Our research will provide an easy-to-use system that enables ongoing monitoring and evaluation.

And, importantly, many of the current studies only address one of the issues such as prediction, classification or routing. This study will fill this gap by creating an integrated, smart complaint management system that automatically classifies a complaint into various government departments, prioritises the complaints through Machine Learning algorithms, allows citizens to vote on the complaints, automatically monitors the complaints with a unique minister, assigns an agent to the complaints, and tracks the complaints.

1.4. Research Objectives

1.4.1. Main Objective

The main objective of this research is to build GovConnect, an intelligent web-based system which prioritises and categorises citizens' complaints into different government departments by using Machine Learning algorithms. It uses modern Artificial Intelligence (AI) technologies such as Retrieval-Augmented Generation (RAG) and a fine-tuned Large Language Model (LLaMA 3) to efficiently process and classify unstructured complaints. The smart system prioritises and handles the most critical complaints. The system also facilitates government processes by enabling ministers to review and route complaints to appropriate agents, and tracking complaint status in real-time for greater transparency and accountability. Automating the complaint resolution process will help enhance the efficiency, quality and responsiveness of government-citizen interactions and provide better service delivery to citizens.

1.4.2. Specific Objectives

Knowledge Base Creation: Build a structured knowledge base of government departments, descriptions, keywords, synonyms and examples of complaints to help with the classification process

Complaint Classification Model: Design a smart complaint classification model using RAG and LLaMA 3 to classify complaints and send them to the right government department based on the content and context

Text Analysis: Examine unstructured complaint information, such as informal or vague descriptions, to classify complaints

Web Application Development: Developed an intuitive web-based application using the MERN stack, Vite and TailwindCSS for citizens to lodge and manage complaints, making it accessible and easy to use

Minister Dashboard: Create a dashboard for ministers to view and manage complaints, see complaints specific to their department and assign them to agents to resolve

Agent Management: Implement a system for allocating complaints to agents based on department and district to ensure that complaints are assigned to the right agent

Complaint Status Tracking: Implement a status tracker that enables agents to update the status of complaints, with real-time status (e.g., pending, in progress, resolved).

Real-Time Monitoring: Allow citizens and government to monitor the status of complaints in real time, promoting transparency and accountability

System Evaluation: Evaluate the system's performance, accuracy, user-friendliness and practicality

Priority Sorting: Use a Random Forest Regressor to prioritize complaints in a list, based on urgency and citizen feedback, to prioritise action on critical complaints

Community Voting: Integrate a citizen voting system to help determine complaint urgency and make the system more democratic and equitable

Complaint Handling Workflow: Create a workflow for the system to classify, prioritise, track and manage complaints, with updates provided to ministers and citizens

2. METHODOLOGY

2.1. Understanding the Key Pillars of Research Domain

GovConnect rests upon some fundamental technologies/principles: Natural Language Processing (NLP), Retrieval Augmented Generation (RAG), Large Language Models (LLMs) and Machine Learning (ML). Combined, these

three technologies give an intelligent system for complaints management by automatically classifying, routing, and prioritizing complaints. More specifically, the Machine Learning-Based Prioritization pillar is particularly important. This pillar uses Random Forest Regression models to predict the priority score of each complaint, so as to quickly prioritize urgent cases. The model predicts this by using certain factors like severity of the complaint, priority given by the citizens, and engagement levels with citizens etc. And as compared to traditional methods of managing complaints, the system achieves more accuracy and better efficiency in the same.
Another significant column is the Text-Based Complaint Classification which allows the system to auto-categorize the complaints to the designated government department. Utilizing NLP and RAG, GovConnect can comprehend text-based complaints even in a context where there is informal, imprecise and incomplete information. This is needed to automatically categorise and respond to each complaint efficiently and correctly in a dynamic environment without any human intervention. The NLP-based classification helps in routing complaints to the correct governmental entity, whereas the RAG retrieves the relevant information to support the classification based on each department.

Lastly, Citizen Participation serves as another critical column. This column allows citizens to vote the urgency level of the complaint which ensures that the prioritization is a fair process reflecting public opinion. The system combines AI technology with citizen participation to make GovConnect responsive to the demands of public and ensuring that urgent issues are attended without delay. This is also supplemented with workflow management that permits the minister to watch over the system, assign the complaints to the agent, and also track its resolution process, thus being held responsible.

2.1.1. Data Collection and Dataset Preparation

This data was collected during the visits to government offices at Kaduwela, Athurugiriya and Colombo municipal councils. Discussions with officers there helped me understand the current manual complaint handling mechanism and provided details about different departments categorized as transport, health, environment & public nuisance, waste management, education, water supply and electricity & power. Officers were happy to help and provided examples of real complaints such as 'Road is completely damaged and vehicles cannot pass', 'No water supply for three days' to build the knowledge base in the form of department names, description, keywords, synonym and examples of complaints stored in the knowledge.json file. This data was converted into vector embeddings using the embedding model and indexed in index.json for RAG. This was done to increase the accuracy of RAG and match complaints based on meaning and not just on the key words used. A dataset was prepared in the CSV file including complaint title, number of votes, priority(HIGH/MEDIUM/LOW), status of the complaint(PENDING/IN PROGRESS/RESOLVED/REJECTED) and urgency score which represents the critical nature of the complaint. The urgency score is used as the dependent variable for the machine learning model as it is based on all other variables mentioned before to define a complaint as urgent or not.

2.1.2. Complaint Submission Process

Citizens access GovConnect through a Web Interface and report issues by typing in necessary information such as the title, description, priority, and location of the complaint. The system records the complaint and puts it into the MongoDB database and marks the status of the complaint as PENDING.

2.1.3. Feature Engineering

Feature Engineering is a crucial element in prioritizing complaints within the GovConnect system. A complaint is described by key features like the vote count, the priority (defined by citizens with HIGH = 4, MEDIUM = 2 and LOW = 0), and other characteristics to determine the rank. These features are used to calculate the

raw score based on: Raw Score = Votes + Priority Bonus and feed it as an initial variable to the Random Forest Regressor model, which outputs the urgency score of each complaint.

2.1.4. Machine Learning Model and Complaint Handling Workflow

Once all complaints have been submitted and classified, the urgency scores of complaints are calculated based on their respective raw scores through the use of Random Forest Regressor which is used for predicting the urgency scores from the raw scores. Then, complaints are sorted with the highest urgency at the top. Complaints with highest urgency get processed before others so that the officials can address urgent cases. In addition, citizens can use voting to the influence the urgency scores further; citizens can vote up on the issues they think deserve the highest urgency so that this leads to re-evaluation of the urgency scores on a continuous basis.

2.1.6 Large Language Models (LLMs)

NLP is a vital part in GovConnect system. Citizens generally submit their complaints in natural text. As the complaints written by citizens have some features like poor spelling, use of colloquial words and vague structure, which would make classification of complaints difficult. For an example, if a complaint says "road broken, vehicles cannot go" then we cannot identify the particular department involved, but the complaints are relevant to roads and transportation. Hence NLP technology is used to clean and normalize the free text of the complaints and remove redundant characters. These clean contents are converted into numerical forms by using embedding methods and these numbers will help the system to recognize the meaning of complaint to classify in the correct department. This leads to higher accuracy of classification when real input data is considered.

Intelligent decision making within the GovConnect system is powered by Large Language Models (LLMs), specifically LLaMA 3. These Large Language Models can usecontext to infer meaning and make decisions based

on information provided. GovConnect, instead of using the Large Language Model for classification purposes, takes a hybrid approach and uses the Large Language Model as a secondary decision maker when the retrieval stage is not confident enough. The top candidate departments determined by the retrieval stage are fed into the LLaMA 3 model, which chooses which of the departments is the most likely. By taking this more constrained approach with the use of LLMs, less computational power is used, responses can be relied on more, and wrong responses can be avoided due to the reasoning power of LLMs.

2.1.7 Information Retrieval and Similarity Matching

Another essential component in the RAG based classification system is the use of Information retrieval techniques. The complaint texts and department knowledge texts within GovConnect are first converted to vector representations. The system uses an embedding-based similarity matching process to measure the extent of similarity between the complaint texts and the knowledge base. The degree of similarity between them is computed based on mathematical distance metrics. It includes both the semantic similarity and keyword matching to help the system identify complaint even with synonyms and similar phrase in complaint texts, for example a complaint on "power cut" can be identified through both the key words and similarity measure thereby aiding in department classification and also a confidence threshold can be set in to avoid irrelevant departments from being presented.

2.1.8 Web-Based System Engineering

GovConnect has been implemented as a web based application. This ensures ease of use, scalability and accessibility of the system. The web based application has been implemented using the latest stack of technology using React for the front end, Node.js and Express for the back end and MongoDB for the database. This allows seamless integration of the system allowing complaints to be submitted, departmental categorizations and then monitoring and tracking complaints dynamically. Being a web based system allows seamless integration of the machine learning model to ensure classification

and prioritization of the complaints is done without manual interference while ministers and agents handle the work flow.

2.1.9 Role-Based Access Control and Security

To protect the system and preserve the confidentiality of information, GovConnect has a role based access control (RBAC) implementation, which permits users only to access the parts of the system that are appropriate to them. The authentication to the system is handled via JSON Web Tokens (JWT) which secure session management and prevents API endpoints from unauthorised access. Citizens, ministers and agents have a variety of role based permissions which only give them access to the parts of the system relevant to them.

2.1.10. Retrieval-Augmented Generation (RAG)

One of the key features that makes up the GovConnect system isRetrieval-Augmented Generation(RAG), which merges the functionality of information retrieval systems with that of generative models. Conventional machine learning models may be dependent solely on their training data, while standalone LLMs might present classifications without a knowledge base to anchor it. Retrieval-Augmented Generation enhances the classification accuracy by not only retrieving department-related information from a structured knowledge base but by coupling that with a language model to produce domain-specific classifications. The structured knowledge base is generated using information regarding government departments such as a department's description, its keywords, synonyms of its name, and some examples of a complaint related to the department stored in a JSON file. Once a complaint is logged into the system, similarity matching is employed to find department-related information closest to that complaint; the retrieved department-related information is then used with the language model for a more accurate classification.

2.2. GovConnect Complaint Classification and Assignment Approach

The GovConnect system was established in order to automate and simplify the handling of complaint requests by prioritizing, classifying, and routing of

complaint requests efficiently. In order to manage complaint requests, the methodology begins by gathering the complaint's relevant attributes from government offices, which includes complaint title, number of votes, priority (HIGH, MEDIUM, LOW), status (PENDING, IN PROGRESS, RESOLVED, REJECTED), and urgency score of complaint. Then, the complaint attributes are organized to create a knowledge base that will then be embedded and indexed for use in the RAG process, so that the complaint can be correctly classified into the appropriate government department according to its context.

Second, GovConnect incorporates an LLM-powered classification component that provides suggestions on appropriate departments in context to the respective complaint. This component makes classification an automated process, through machine learning, that decreases the human effort. Upon classification, the complaint enters a workflow wherein ministers can be assigned and assign the complaint to agents in relation to the department and the district. The agents are then accountable for the update of the complaint status.

One important component is the Random Forest Regression model, which computes the urgency of each complaint by considering the priority selected by citizens and the number of votes. These values determine the ranking of the complaints in order from highest urgency to lowest urgency so that high-priority issues are addressed promptly. The complaint list ranked from highest urgency to lowest is presented on the ministerial dashboard where tasks can be monitored. GovConnect system allows for effective monitoring and assignment of tasks, incorporating machine learning, citizens' input, and automation into a coherent and workable application.

2.2.1 Data Collection

In order to collect data for the above study, responsible officers from the government offices located at Kaduwela, Athurugiriya and Colombo Municipal Council were approached and discussed about the present manual complaint handling mechanism. The information from the officers stated that

complaint handling system is still a manual process where the officers read the complaints and allocate the relevant departments for those complaints depending on their experience and expertise. The officers also named the relevant departments in which the complaints could be categorized namely.

The usual complaint samples, along with practical problem patterns concerning each department was gathered through this interaction. The examples that came about were 'Road is completely damaged and vehicles can not move', 'Hospital emergency ward full with patients', 'Garbage has not been collected for two weeks, 'No water for 3 days in house' and 'Every evening there is a power cut', which were typical for each department. These were used to build up a structured knowledge base for RAG in classifying the complaint effectively. The set differs from any public domain data because this is based on the domain knowledge of the local government and so, the recommendations will make sense in practice.

The prioritization component of the system also uses the results of the interview with officers, which showed that priorization is also being done manually. It is on officers experience that urgency is attributed to each complaints.The extracted and the interview information are combined to form a dataset composed of complaint titles, priorities, votes, status and urgency. This dataset allows the system (GovConnect) to be trained with real-world local data, increasing its accuracy and effectiveness.

2.2.2 Knowledge Base Preparation

In order to create an intelligent complaint classification, and routing process in the GovConnect system, an efficient and well-structured knowledge base had to be built. To this end, actual complaint information collected from government departments situated in Kaduwela, Athurugiriya, and Colombo Municipal Council was analyzed. This analysis allowed gathering some interesting information about the current process of complaint handling and

classification into categories. For example, some of the categories into which complaints could be classified included transport, health, environment & public nuisance, waste management, education, water supply, and electricity & power.

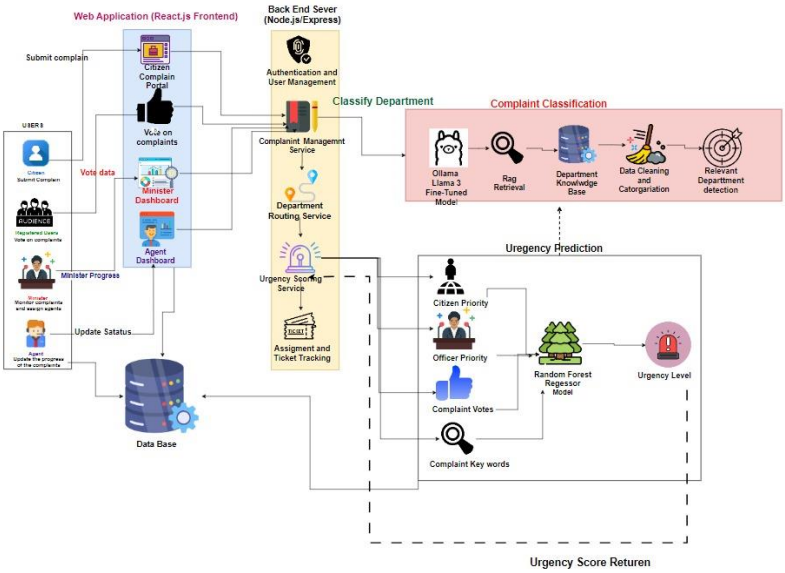
An essential part of preparing the knowledge base involved building a well-structured dataset which included complaint title, vote count, priority (high/medium/low), complaint status (pending, in progress, resolved, rejected), and urgency score as attributes. The dataset was stored in CSV format and served as a source of training. Here, the urgency score denotes how urgent a particular complaint is, and thus, it serves as a dependent variable in the machine learning model under development. Using historical complaint data, the model was built and trained using the Random Forest Regressor algorithm. Thus, using the model, the prediction of urgency score for a complaint could be made using its votes, priority, and category.

To ensure that the GovConnect classification mechanism correctly classified complaints, the acquired information from the different departments was put into knowledge form by providing not just their name and description but their keywords and example complaints among other pieces of information. This knowledge was then stored in a knowledge.json file, which was crucial to how the Retrieval Augmentation Generation (RAG) classification mechanism operated. RAG is a combination of both information retrieval techniques and language models used for enhancing the classification of complaints by finding semantic matches between complaint descriptions and the provided departmental knowledge rather than keyword matches, thus minimizing the risk of misclassification.

Apart from the knowledge.json file, the complaints data were also converted into vectors using an embedding model. Embedding models take complaint data and convert them into mathematical vectors that represent the semantic meaning of the data. These vectors were then indexed and stored in another file called index.json. The use of vectors in the system enables it to perform semantic matching, which helps the system to retrieve information related to the department regardless of the way complaints have been phrased.

Therefore, a complaint relating to “damaged roads” will be semantically mapped to the Transport Department.

2.3. High-Level System Architecture



The system architecture of the GovConnect platform successfully integrates several elements that cooperate to automate the government-citizen complaint handling procedure. The architecture is divided into discrete levels, each of which is in charge of managing particular functions, guaranteeing an effective information flow, and facilitating smooth communication between users and governmentagencies.

React.js is used in the platform's frontend development, offering a dynamic and interactive interface for users to file complaints. While ministries and agents have access to dashboards to monitor and address problems, citizens can readily file complaints. Details from complaints submitted by citizens are processed in the backend for routing and classification.Assigning complaints to the proper agents and monitoring their progress are made possible by the Minister function. Agents are in charge of managing the complaints that are allocated to them, keeping their statuses up to date, and resolving problems.All essential functions are managed by the backend server, which is powered by Node.js and Express.js. This involves handling complaints and utilizing JWT-based user management for authentication. Additionally, the backend incorporates RAG and LLaMA 3 models for complaint classification, automatically allocating complaints to the relevant department

according to the substance of the complaint. In order to handle complaints according to urgency, the Urgency Scoring Service also use Random Forest Regressor to determine each complaint's priority.

Classification and urgency prediction heavily rely on AI models. By examining the complaint content and comparing it with department records kept in the knowledge base, RAG and LLaMA 3 collaborate to guarantee appropriate department classification. Additionally, the system uses Random Forest Regressor to forecast urgency while accounting for user votes, officer priority, and citizen priority.Lastly, complaint records, user profiles, departmental data, and agent assignments are stored in the database, which is run by MongoDB. This ensures that the system runs smoothly even as the number of complaints increases by enabling effective data retrieval and handling.

All things considered, the architecture of the GovConnect platform guarantees that complaints are resolved quickly, effectively, and precisely, minimizing manual labor and improving the citizen-government engagement process.

2.4. Functional and Non-Functional Requirements

2.4.1. Functional Requirements

Functional Name	Input	Processing	Output
Citizen Registration and Login	User details such as name, email, password,	System validates input, encrypts password, stores user details, and authenticates using JWT	Secure user account creation and login access
Complaint Submission	Complaint title, description, phone number, address, district, priority, and optional department	System validates complaint data and records it in the database	Complaint stored successfully in the system
AI Department Suggestion	Complaint title or complaint text entered by citizen	RAG system retrieves relevant knowledge from	Suggested department returned to the

		knowledge.json, compares embeddings, applies hybrid score, and selects the most suitable department	citizen
Manual Department Selection (Optional)	Department selected manually by citizen	System bypasses AI suggestion and directly uses selected department	Complaint routed to selected department
Knowledge Base Matching	Complaint text and department knowledge data (description, keywords, synonyms, examples)	System checks semantic similarity and keyword relevance against indexed knowledge	Top matching departments identified
Complaint Storage and Routing	Complaint data with final department	System stores complaint in MongoDB under the matched department	Complaint available in relevant department workflow
Minister's Complaint View	Department-based complaint request from minister	System filters complaints according to the minister's assigned department	Department complaint list displayed in minister dashboard
Agent Filtering and Assignment	Complaint department and district, available agent data	System filters agents by department and district; minister	Complaint assigned to the most suitable agent

		selects one for assignment	
Agent Complaint Handling	Assigned complaint and updated status	Agent reviews complaint and updates status such as Pending, In Progress, Resolved, or Rejected	Updated complaint status stored in database
Complaint Tracking	Complaint ID and user role	System retrieves complaints, progress and status	Citizens, ministers, and agents can monitor complaint progress
Admin User Management	Minister, agent, and department details	Admin creates departments, registers ministers and agents, and assigns ministers to departments	Updated user and department records
Role-Based Access Control	Authenticated user token and role	System checks user role permissions before granting access to APIs and dashboards	Controlled access for admin, citizen, minister, and agent
Complaint Submission	Complaint title, description, priority level, district, address, contact details	System collects complaint details from citizen form and stores them in the database	Complaint successfully stored with default status as **Pending**
Complaint Data Preparation	Complaint dataset with votes, priority, status,	Dataset is organized and prepared for	Structured dataset for urgency

	urgency score	machine learning model training	prediction
Urgency Score Prediction	Complaint votes and priority level	Compute raw score and apply **Random Forest Regressor** to predict urgency score	Predicted urgency score for each complaint
Complaint Ranking	Predicted urgency scores	Complaints are sorted in descending order based on urgency score	Ranked list of complaints from most urgent to least urgent
Citizen Voting Mechanism	Citizen vote for a complaint	System records vote, prevents duplicate voting, and updates complaint importance	Updated vote count and priority influence
Minister Dashboard	Ranked complaints by department	Display complaints in urgency order and allow ministers to review them	Minister can identify and manage urgent complaints
ML API Integration	Complaint records from backend	Backend sends complaints to Flask ML service for prediction	Urgency scores returned through API

2.4.2. Non-Functional Requirements

NFR	Testable NFR
Performance	The system should return department suggestions within a few seconds under normal load.
Scalability	The system should support a growing number of

	complaints, users, departments, ministers, and agents without major performance degradation.
Availability	The system should remain accessible during normal working hours with minimum service interruption.
Reliability	The complaint classification and routing process should produce consistent results with minimal system failure.
Usability	A citizen should be able to submit a complaint with minimal effort using a simple and understandable interface.
Maintainability	The modular architecture should allow new departments, keywords, synonyms, and complaint examples to be added without major code restructuring.
Compatibility	The web application should work correctly on common modern browsers such as Chrome, Firefox, Edge, and Safari.
Security	The system should enforce JWT-based authentication and role-based access control to prevent unauthorized access.
Accuracy	The department classification component should maintain high accuracy using the RAG + LLM hybrid method and should correctly classify most real complaint cases.
Auditability	The system should preserve complaint records, assignments, and status updates so they can be reviewed when necessary.
Data Integrity	Complaint, user, and department data should be stored consistently without duplication or corruption.
Flexibility	The system should support future extensions such as

	adding new complaint categories, departments, and workflow features.
Performance	System responds within 2 seconds under normal load.
Scalability	Supports many users and complaints without performance loss
Availability	System remains accessible during working hours.
Reliability	Ensures consistent ranking and works with fallback if ML fails.
Usability	Users can easily submit, vote, and track complaints.
Maintainability	Modular design allows easy updates without failure.
Compatibility	Works on modern browsers (Chrome, Firefox, Edge, Safari).
Accuracy	Produces reliable urgency scores for correct ranking.

2.5. Social, Security, and Ethical Considerations

2.5.1. Social Impact

GovConnect: An Online system for complaint handling that offers better communication between citizens and government by consolidating a complaint handling platform online. Old complaint systems are tedious as there is a lack of coordination and a manual system, which causes inefficiency in resolving and replying to a complaint. In contrast, GovConnect utilizes intelligent classification mechanisms in order to effectively assign a complaint to the appropriate authority thus reducing time.

The real-time tracking feature enables citizens to check on the progress of their complaints, thus increasing public trust in government service providers. The voting feature also allows citizens to convey the importance and urgency of their complaints, prioritizing the more pressing issues. The interface, being user-friendly and simple, further ensures that the general population can easily utilize the system for their service needs without prior technical expertise.

2.5.2 Security Considerations

Security is an essential part of the GovConnect system as it involves handling personal and complaint data. To implement this, it uses JSON Web Token for safe authentication and authorization. This means that users who are authenticated, will be allowed to certain functions on the system and users that are authorized can access particular functions. It is uses role-based access control to limit users who are only able to view certain parts of the system depending on the role they have on the system. This means only citizens will have access to citizen's complaint details; ministers would only be able to view ministerial reports; an agent can only view agent related data; an administrator will have full access to all data within the system. Sensitive data like the password is kept hashed so that it is kept safe in case of a data breach. In terms of input validation and sanitization techniques; injection attacks, these methods can be used to ensure that there is no data manipulation.

2.5.3. Ethical Considerations

Several ethical concerns have been identified with the use of Artificial Intelligence (AI) in GovConnect. Firstly, concerns about accuracy and fairness arise as the system classifies and prioritizes citizens' complaints objectively. To guarantee that a citizen can make the right decision about where to direct the complaint, the system has been implemented with a fallback feature allowing the user to make a choice in departments if the automated choice is not ideal. Secondly, attempts have been made to prevent a bias in the machine learning model by implementing a structured knowledge

base in which all departments are treated equal such that no one is favored or penalized unjustly. It relies on explicit variables such as the number of votes or priority level, thereby eliminating decisions based on inappropriate or biased decisions, or actions that might cause harm to a citizen. Finally, user privacy is respected as all private information are disclosed to selected users only, while an access control is set for controlling sensitive data usage, and accountability is maintained by logging each step of complaint processing, making it easy to find a particular step when required.

2.5.4 Technical Standards and Considerations

The GovConnect system was implemented using good coding standards to achieve long-term maintainability and scalability of the system. The application was developed using the MERN stack: MongoDB, Express, React, and Node.js and a machine learning API that utilized Python to predict the prioritization of each complaint. It was coded using structured and modular programming conventions by adhering to object oriented programming (OOP) methodology. This ensured readability and maintainability of the code, along with clear comments and naming conventions. The system's constraints include: the machine learning algorithm used to predict priority has been trained on limited data and could benefit from being trained on more data, the prediction algorithm currently uses a raw score derived from the values of Votes and Priority only, and this could incorporate other data, such as Type of Complaint and Location of Complaint, in later revisions of the program. The system may also face responsive design issues while being accessed on a mobile phone, and the functionality of the web application depends on the status of the python backend, although a fallback plan was developed.

2.6. Testing Strategy

The testing strategy of the GovConnect system was designed to evaluate whether the developed platform can correctly classify, route, and manage citizen complaints using the proposed AI-based complaint classification pipeline. The main focus of the testing process was to validate the

performance of the RAG-based department classification model, LLaMA 3 fallback decision process, complaint routing mechanism, role-based access control, and system authentication process.

The testing was carried out using a set of real-life complaint test cases that represent different government-related issues such as transport, health, education, environment, public safety, road maintenance, and other public service concerns. When a citizen submits a complaint and selects the AI suggestion feature, the complaint text is first converted into an embedding vector using the embedding model. This vector is then compared with the pre-processed department knowledge base vectors using semantic similarity.

In addition to semantic similarity, keyword matching and synonym matching were also tested to improve classification accuracy. The system calculates a hybrid score by combining semantic similarity and keyword-based relevance. A minimum confidence threshold of **0.45** was used to decide whether the suggested department is reliable. If the top-ranked department achieved a hybrid score greater than or equal to this threshold and existed in the database, the system accepted it as the final department suggestion.

For ambiguous complaints where the hybrid score was below the required threshold, the system used the LLaMA 3 model through the Ollama interface. In this stage, the complaint text and the top candidate departments retrieved by the RAG system were passed to the LLM. The LLM then analyzed the complaint context and selected the most suitable department. This fallback mechanism was tested to ensure that unclear or complex complaints could still be categorized accurately.

After classification, the system was tested to verify whether the complaint was correctly stored in the database and routed to the relevant department dashboard. The routing process was also tested based on department and district. For example, if a complaint related to transport was submitted from the Matara district, the system should allow the minister to assign it to a relevant Transport or Road Service agent in the Matara district.

The testing strategy also included validation of authentication and role-based access control. The system uses JWT-based authentication to ensure secure

login and session handling. Different user roles such as citizen, admin, minister, agent, and police were tested separately. Citizens were tested for complaint submission and viewing their complaint status. Admin users were tested for registering ministers, agents, and police users. Ministers were tested for viewing department complaints and assigning complaints to agents. Agents were tested for updating complaint status. This ensured that each user could only access the functions assigned to their role.

2.6.1. Complaint Classification Accuracy Testing

The accuracy of the complaint classification system was evaluated using **39 complaint test cases** collected from real-life complaint scenarios. Each complaint was processed through the RAG classification pipeline, where the complaint embedding was compared with the embedded knowledge base. The final department was selected based on the hybrid score and the confidence threshold of **0.45**.

Out of the 39 test cases:

Result Type	Number of Test Cases
Correctly classified complaints	34
Incorrectly classified complaints	5
Total test cases	39

The accuracy was calculated as follows:

$$Accuracy = \frac{Correctly\ Classified\ Complaints}{Total\ Test\ Cases} \times 100$$

$$Accuracy = \frac{34}{39} \times 100 = 87.18\%$$

The obtained accuracy of **87.18%** shows that the proposed complaint classification system performs effectively in identifying the most relevant department for citizen complaints. The result also indicates that the hybrid RAG and LLM-based approach improves classification reliability, especially when complaints contain unclear wording, synonyms, or indirect descriptions.

Overall, the testing strategy confirmed that the GovConnect system can support real-time complaint classification and routing. The system successfully combines a knowledge base, embedding-based similarity comparison, keyword and synonym matching, hybrid confidence scoring, and LLaMA 3-based fallback decision-making. In addition, the testing process verified that complaints are correctly routed to the relevant department and managed through secure role-based access. Therefore, the testing results prove that the proposed system is suitable for improving the efficiency, accuracy, and transparency of government complaint handling.

2.7. Tools and Technologies Used

Frontend Technologies

The frontend part is created by React.js, this allows to have a component-based, reactive and flexible interface that adapts according to user's role and complaint data. Vite is used as build tool, in order to get the code built faster and efficiently rendered. Tailwind CSS is used for the design of the responsive, user-friendly interface, to create the same and accessible design across all user's page: citizens, ministers and agents. Axios is used to make connection between frontend and backend in order to easily fetch data from backend APIs.

Backend Technologies

The backend of the application is implemented in Node.js and Express.js. Node.js gives us a powerful and performant environment for asynchronous operations and concurrent requests. Express.js is a Node.js web application framework which helps us create RESTful API endpoints for the functionality like complaints submitting, authentication, status update, assigning agent etc. We have chosen MongoDB as the database management system to persist our complaint data, user information, voting etc. Mongoose is an Object Data Modeling (ODM) tool which we used to make it easy to interact with MongoDB from the backend.

Machine Learning Technologies

Another module is a machine learning model. It is created by using python language and it uses scikit-learn library. Random Forest Regressor model accepts all different complaint parameters like count of vote, priority of the complaint etc and based on those parameters, it determines an urgency value of a complaint. Urgency value helps to arrange complaints according to their priority, highest priority at first. This model is installed in the system to automatically identify priority of a complaint.

Data Processing Tools

These are the libraries used in the machine learning pipeline to process and handle the data. Pandas and Numpy are used for data wrangling, cleaning and transformation to have a usable dataset for the machine learning model. Joblib is used for serializing the models such that the trained model can be save to load during the training phase and later during the prediction phase.

ML Deployment and API Integration

For deployment of the machine learning model the Flask framework is employed, which forms a python based API for scoring and training the model. This would serve as an intermediary service which could receive complaints from the backend and send it to the machine learning service for classification and getting theurgency score. The communication between the Node.js backend and the python ML service can be done through axios.

Fallback Solution

A backup was set in place so that the system doesn't stop working if the Python ML service fails. This solution uses JavaScript and so ensures that essential complaint management functions aren't hindered by any failures.

Security Techniques

The GovConnect system utilizes JSON Web Tokens (JWT) to authenticate that only users who are authorized can access certain functions of the platform, therefore protecting system components from being accessed unlawfully. RBAC (Role-Based Access Control) is used in the GovConnect system to ensure that roles can only access certain parts of the system (e.g. Citizens only access the 'citizen' part of the system, ministers only access

'minister' etc.).

Integration of the Overall System

The GovConnect system is built upon the MERN stack by merging backend, frontend, and machine learning functionalities into one platform to manage complaints with real-time updates and tracking. Moreover, the system has a modular design to ensure its ability for further expansion, for the reason that the application is feasible to scale up to cloud.

4.RESULT AND DISCUSSION

4.1. Result

4.1.5 End-to-End Workflow Results

The whole complaint handling work flow has been tested in order to confirm whether all the functionality of the system has been integrated to it.The complaint handling work flow system includes the following steps: The citizen registers a complaint, which is then directly integrated to the system as per the RAG based AI module and stored into the data base. The complaint will appear in the minister's portal and the corresponding minister will get the alert. The complaint will be assigned to the appropriate agent as per the department and district. The assigned agent will be updating the complaints according to its stage i.e.,pending,in progress,finished. Each and every stage has been tested. And no variation from the current workflow. The complaints were also made towards the relevant departments, agent being appropriately assigned to the complaint according to the departments and districts.

4.1.6 System Performance Results

The performance of the system was monitored by looking at the response time of the key backend API calls. All the main API calls of the GovConnect system had the following response times (these are approximate times): The creation of a department finished at around 240ms, agent registration at around 513ms, minister registration at around 644ms, admin login at around 226ms. From these response times, it can be seen that the backend services and database calls together with the authentication process seem to perform in an efficient way for the given load conditions.

4.1.7 Testing Accuracy of Complaint Classification System

4.2. Research Finding

The performance evaluation of GovConnect system has shown a variety of results with respect to the efficacy of AI based complaint classification as well as system integration as a whole. The findings prove that a combination of RAG with LLama 3 will perform efficiently and scalable for automating government complaints.

It is important to note that the overall accuracy for this system is 87.18%, with 34 of the 39 complaints classified correctly. The results indicated that the combination of semantic similarity, keyword detection, and synonym replacement is successful in processing natural language complaints. Using an embedding-based semantic match increases the understanding of the complaints, regardless of whether they are short, ungrammatical or written in a colloquial style.

The 0.45 confidence threshold mechanism has also proven vital to the classification reliability as it filters out poor confidence decisions and retains high accuracy. This has also been combined with the LLM fallback mechanism to accommodate fuzzy cases, where only the semantic similarity of the documents will not suffice.

The system also provided an efficient real-time implementation with classification of the complaint, mapping it to a department, and initiating a workflow all performed in less than a second and in one flow. All of the critical APIs returned response times that averaged less than 1 sec and the implementation proved itself to be effective for real time usage.

It has been noted from the above results that workflow automation based on the roles of persons (ministers, agents) has proved itself to be beneficial by directing complaints to respective ministers and assigning complaints to agents by department and district. It shows the system can be used not only for classification but also for full complaint management.

Metric	Value
Total Test Cases	39
Correct Pedictions	34
Incorrect Predictions	5
Accuracy	87.18%

Table 4.2.1: Classification Performance Summary

4.3. Discussion

4.3.1. Effectiveness for AI-Based Classification

This experiment confirms that the GovConnect can successfully classifying citizen's complains through hybrid AI. Comparing with usual classification techniques using keyword matching, this system use the semantic embedding for complaint understanding. So the system can classify correctly even though there is no keyword presented or uses another word form.

The resulting 87.18% accuracy is very good and more importantly that the system was built with real life data i.e. Complaint data and not simulated data, we can be confident that the RAG combined with LLM method is a valid method for the purpose of e-governance.

4.3.2 Impact of Hybrid Scoring Mechanism

The use of hybrid scoring function also helps to achieve high accuracy in classification. Semantic similarity coupled with keyword and synonym matching consider the complaint text on more than one aspect during classification process.

Such methods are highly relevant to tackle with::

- Informal complaint texts
- Short text
- Wording variations

Another measure that increases the prediction reliability is to use a confidence threshold value. With such a threshold we do not accept only good matchings. This helps to avoid false positives and generally increases the efficiency of the system.

4.3.3 Role of LLM in Handling Ambiguity

Perhaps the most valuable part of this system is the use of LLaMA 3 as a fallback when the hybrid score is too low and contextually reasoning the appropriate department through the LLM. This vastly increases the system's capacity to address:

- Vague complaints
- Multi-context issues
- Overlapping department categories

As another example, "complaints about a vehicle accident could relate to transport or health". The LLM can clarify this ambiguity by taking the entire meaning of a complaint into account.

4.3.4. Workflow Automation and System Integration

In addition to the classification task, the GovConnect system integrates various elements together into one process flow. It demonstrates that complaints were directed automatically to the correct department, examined by ministers and distributed to the correct agents (depending on district and type of service). The overall process automation minimizes the amount of manual work and efficient in dealing with complaints and each complaint had followed a clear process from being received and investigated.

4.3.5 System Performance and Scalability

Overall, the results of system performance show that all API requests are fulfilled with an acceptable amount of time. The most part of all API request completes with an execution time less than 1 second. Therefore, system has a good capacity for fulfilling real-time requests.

The system is designed with the MERN stack which will make the system scalable when more and more users and complains happen. Modular design allows us to add more features and updates easily in the future.

4.3.6 Limitations and Error Analysis

Even though accuracy was good, the system was incorrect for 5 out of 39 cases tested, this was mainly due to the following problems:

- The complaint text was too ambiguous
- Departments had too many similar keywords
- There was no distinct clue to guide which department to assign to the complaint

An example of the system being wrong is the complaint about an accident being classified as Health rather than Transport because of the shared context.

4.3.7 Adaptability and Real-World Applicability

GovConnect shows adaptability to different formats of complaint, i.e. Informal text, or partial descriptions, which are both crucial for real applications in which users may not strictly follow formal inputs.

This proposed system does not need constant retraining unlike standard machine learning systems, but rather its knowledge can be continuously and dynamically updated as new types of complaint arise, or government organization and relationships change.

4.3.8 Comparison with Traditional Systems

Compared to the traditional machine learning approaches, there is no need of explicit training, validation and retraining for the proposed system GovConnect. Classification is performed by dynamically combining a knowledge-base that can be ever-expanding and semantic retrieval mechanism. This provides much higher flexibility and scalability by adding new complaint types without the need to retrain a model.

5.CONCLUSION

The design of the GovConnect system is an example where AI is utilized to better government-citizen interaction through an intelligent complaint handling system. The aim of the current work is to build a system which can classify citizen complaint to appropriate departments with advanced AI, and combine with a total workflow starting from submitting complaint, directing it, supervising it and allocating it to agents.

The findings of the system evaluation shows that the proposed system has effectively solved the limitations of previous complaint management systems. The addition of Retrieval-Augmented Generation (RAG) on the LLaMA 3 model makes it easier to comprehend the contextual semantics of a complaint instead of a mere keywords match system, which enhances the classification accuracy of the system and allows it to tackle actual complaints as input might be informal, lack some information, or unstructured. The attained 87.18% of classification accuracy shows that the system is reliable in predicting the right departments for most complaints.

Another successful result of the study is the implementation of a hybrid classification mechanism that incorporates semantic similarity, keyword matching, and synonym matching. These approaches increase the system's robustness and make sure that different language characteristics of complaint texts are taken into account while classifying. It is worth noting that a confidence threshold mechanism is used to improve the reliability of the classification decisions by removing those with low confidence. When the classification confidence is not high enough, the system takes the LLaMA 3 model as a backup, which increases the system's ability to resolve challenging or ambigous complaints.

Besides classification the GovConnect system offers a full end to end complaint management solution. It allows the citizens to post their complaint using a simplified UI, then automatically redirects the complaint to the corresponding department. Ministers can keep track of complaints on special dashboards and assign agents according to the department and location of the

complaint. Agents then follow up with the citizens and update the status of the complaint all through the complaint lifecycle.

This performance result from the system execution also illustrates the efficiency and scalability of the proposed solution. The execution response time for the backend system, where complaint classification, department allocation, and user authentication takes place, are within reasonable bounds, i.e., less than 1 second for each operation. Thus, the system is capable of carrying out interactions real-time and hence fit for real-world deployment where massive users and complaint volumes will exist.

However, despite the positive outcomes, there are some limitations that were observed during the evaluation process. Misclassifications of some of the complaints are evidence of the difficulties in processing information that can be classified under more than one category. For example, complaints that touch upon several different areas, such as health and transport, can cause classification errors due to common contextual factors. Also, the efficiency of the proposed solution using the RAG model is highly dependent on the content of the knowledge base, which should be complete and accurate.

However, these limitations do not impact the performance of the system to a large extent. Rather, they point out areas where the system can be improved in the future. One of the advantages of using GovConnect, as a solution that has a modular design, is that changes can easily be made to the knowledge base of the system in order to cope with changing complaints and governmental system

Overall, the presented GovConnect solution effectively illustrates how AI-powered techniques can be utilized in order to make complaint management in the government more efficient and transparent. Through using techniques like RAG and LLMs alongside current web technologies, it is possible to achieve results that can be used to bridge the gap between theoretical research and its practical implementation. Not only will the system allow for complaints classification, but also provide solutions throughout the entire life cycle of the complaint, turning it into a powerful platform for developing e-government solutions.

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